SEPARATED LOCALLY QUASI-FINITE MORPHISMS ARE IND-QUASI-AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19680-19683 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19685-19692, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Flagship example card of the ind-quasi-affine series (dictionary L16109, composition L16116, fpqc-local L16080 all collected); the etale case of the title is subsumed by locally quasi-finite plus separated.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter alternative-definition dictionary for ind-quasi-affine morphisms (card L16109); chapter lemma quasi-finite-separated-quasi-affine (the ZMT quasi-affineness theorem).

Source reference. Original lemma 02LR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12426–12429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A separated locally quasi-finite morphism of schemes is ind-quasi-affine.*

Proof. Let $f : X \to Y$ be a separated locally quasi-finite morphism of schemes. Let $V \subset Y$ be affine and $U \subset f^{-1}(V)$ quasi-compact open. We have to show U is quasi-affine. Since $U \to V$ is a separated quasi-finite morphism of schemes, this follows from Zariski's Main Theorem. See Lemma 02LR. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.